Programming Shared-Memory Parallel Systems - OpenMP

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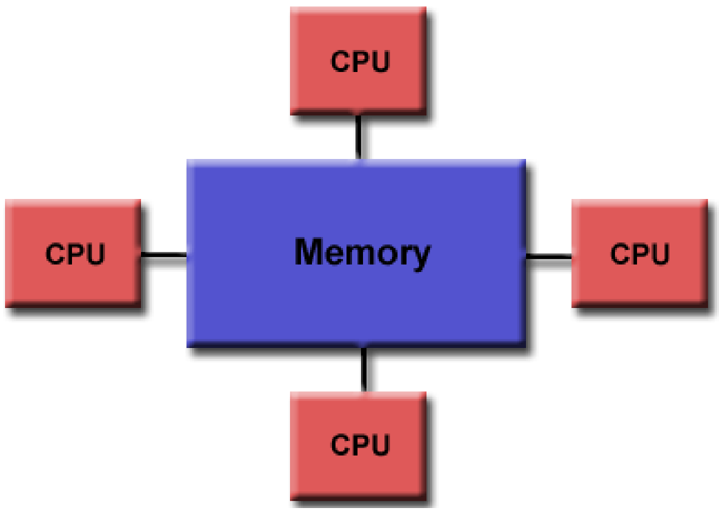
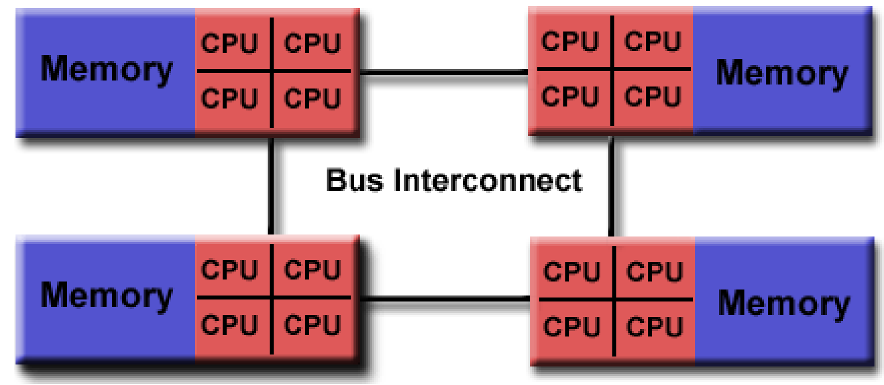
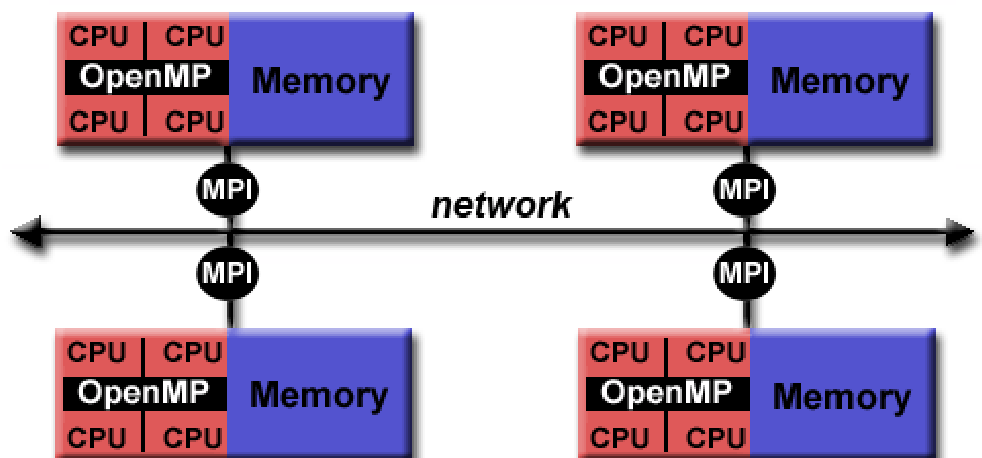
Recall Memory Models

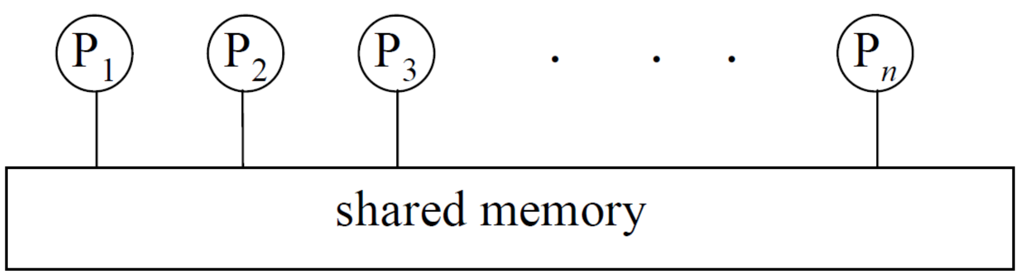






Programming Shared Memory Systems

- The **processes/threads communicate** using **shared memory**
- **UMA/NUMA**
- **Multi-threading**
- You have been using **pthreads** so far.
- Manually implementing all the **parallelization strategy** steps
  - Decomposition
  - Assignment
  - Orchestration
  - Communication/Synchronization



Programming Shared Memory Systems

- **Explicit Parallelism**
  – For example, **pthreads**

- **Programmer Directed**
  – For example, **OpenMP**

OpenMP

- **Programming** of **shared memory systems**
- An **API** for **Fortran** and **C/C++**
  - **Directives**
  - **Runtime routines**
  - **Environment variables**

Lecture credits: https://computing.llnl.gov/tutorials/openMP/

Goals

- **Standardization**
  - **Provide** a **standard** among a **variety of shared memory architectures** (platforms)
- **High-level interfaces** to **thread programming**
- **Multi-vendor support**
- **Multi-OS support** (Unix, Windows, Mac, etc.)
- The **MP** in **OpenMP** is for **M**ulti-**P**rocessing
- *Don't confuse **OpenMP** with **Open MPI**! :)*

Credits: http://its.unc.edu/research-computing.html

Release History

Date	Version
Oct 1997	Fortran 1.0
Oct 1998	C/C++ 1.0
Nov 1999	Fortran 1.1
Nov 2000	Fortran 2.0
Mar 2002	C/C++ 2.0
May 2005	OpenMP 2.5
May 2008	OpenMP 3.0
Jul 2011	OpenMP 3.1
Jul 2013	OpenMP 4.0
Nov 2015	OpenMP 4.5
Nov 2018	OpenMP 5.0

Hello World – pthreads based version

```
#include <pthread.h>
#include <stdio.h>

void* thrfunc(void* arg) {
printf("hello from thread %d\n", *(int*)arg);
}

int main(void) {
pthread_t thread[4];
pthread_attr_t attr;
int arg[4] = {0,1,2,3};
int i;
// setup joinable threads
pthread_attr_init(&attr);
pthread_attr_setdetachstate(&attr,PTHREAD_CREATE_JOINABLE);
// create N threads
for(i=0; i<4; i++)
        pthread_create(&thread[i], &attr, thrfunc, (void*)&arg[i]);


// wait for the N threads to finish
for(i=0; i<4; i++)
        pthread_join(thread[i], NULL);
}
```

... and the OpenMP version

```c
#include <omp.h>
#include <stdio.h>

int main(int argc, char* argv[])
{
   #pragma omp parallel  {
   printf("Hello World... from thread = %d\n", omp_get_thread_num());
   }
}
```

Compilation: $ **gcc -fopenmp hello.c -o hello**

Execution: $ **export OMP_NUM_THREADS=4**

$ **./hello**

Demo: hello.c

Compiling

Intel (**icc**, **ifort**, **icpc**)

- *-openmp*

PGI (**pgcc**, **pgf90**, …)

- *-mp*

GNU (**gcc**, **gfortran**, **g++**)

- *-fopenmp*

OpenMP - User Interface Model

- **Shared Memory** with **thread based parallelism**
- **Not a new language**
- **Compiler directives, library calls,** and **environment variables extend the base language**
  - f77, f90, f95, C, C++
- **Not automatic parallelization**
  - User **explicitly specifies parallelism**
  - **NOTE: Compiler does not ignore** user **directives even if wrong**

OpenMP - Syntax

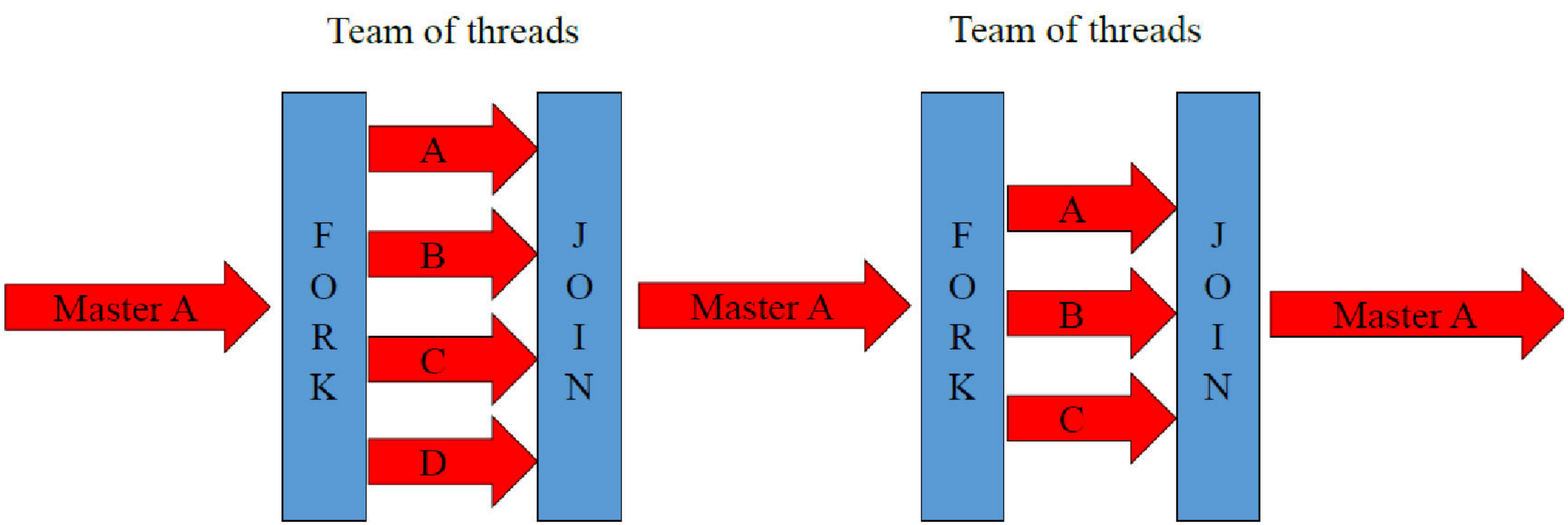
- **Parallelism** is **highlighted** using **compiler directives** or **pragmas**

- **For C and C++,** the **pragmas take the form**:
  `#pragma omp construct [clause [clause]…]`

- **Any compiler** (even if it **does not have OpenMP** support) **can compile** the **program** (with **no parallelism though**)

Fork*/Join Execution Model

- An **OpenMP program starts** as a **single thread** (***master thread***).
- **Additional threads** (***Team***) are **created** when the **master hits a parallel region**.
- When **all threads finished the parallel region**, the **new threads** are **given back** to the **runtime** or operating system



***Not to be confused with fork() system call**

Using OpenMP

- **OpenMP** is **usually used** to **parallelize loops**:
  - **Find most time consuming loops**
  - **Split them** among **threads**

Split-up this loop between multiple threads

```
void main( )
{
    double Res[1000];

    for(int i=0;i<1000;i++) {
        do_huge_comp(Res[i]);
    }
}
```

Sequential program

```
void main( )
{
    double Res[1000];
    #pragma omp parallel for
    for(int i=0;i<1000;i++) {
        do_huge_comp(Res[i]);
    }
}
```

Parallel program

OpenMP Directives

OpenMP - Directives

- **OpenMP compiler directives** are **used** for **various purposes**:
  - **Spawning** a **parallel region**
  - **Dividing blocks** of **code** among threads
  - **Distributing loop iterations** between threads
  - …

sentinel directive-name [clause, ...]

↓ ↓ ↓

#pragma omp parallel private(var)

Supported Clauses for the Parallel Construct

Valid Clauses:

if (logical expression)

num_threads (integer)

private (list of variables)

firstprivate (list of variables)

shared (list of variables)

default (none|shared|private **fortran only**)

copyin (list of variables)

reduction (operator: list)

...

OpenMP Constructs

- **OpenMP constructs** can be divided into **5 categories**:
    1. **Parallel Regions**
    2. **Work-sharing**
    3. **Data Environment**
    4. **Synchronization**
    5. **Runtime functions/environment variables**

OpenMP: Parallel Regions

- You create **threads** in **OpenMP** with "**omp parallel**" pragma
- For example: a **4-thread based Parallel region**:

```
int A[10];
omp_set_num_threads(4);
#pragma omp parallel
{
    int ID =omp_get_thread_num();
    fun1(ID,A);
}
```

Demo: helloFun.c

- **Impl**
- **Each**
- **Each thread executes** the <u>same code</u> within the block

The *parallel* directive

- A **parallel region** is a **block** of **code** that will **be executed by multiple threads**
- **When** (in **serial program**) a **PARALLEL directive** is **found**, a **team of threads** is **created** and **main-thread** (serial execution thread) **becomes** the **master of the team**
- **Master thread** has **id** or **number** **0** (***within that team***)
- **The code is duplicated** and **all threads will execute that code**
- There is an **implicit barrier** at the **end of a parallel region**
- **Master** **thread** **continues execution** after this point

The *parallel* directive

- Some common clauses include:
  - **if** *(expression)*
  - **private** *(list)*
  - **shared** *(list)*
  - **num_threads** *(integer-expression)*

How Many Threads?

- The **number of threads** in a **parallel region** is **determined** by the **following factors**, <u>**in order of precedence**</u>:
  1. **Evaluation** of the **if** clause
  2. **Setting** of the **NUM_THREADS** clause
  3. Use of the **omp_set_num_threads( )** library function
  4. Setting of the **OMP_NUM_THREADS** **environment variable**
  5. **Implementation default:** Usually the ***number of CPUs on a node***

- **Threads** are **numbered** from **0** (master thread) to **N-1**

IF clause

```
#pragma omp parallel if (scalar_expression)
```

- **Execute** in **parallel** **<u>if expression is true</u>**
  - **Otherwise serial execution**

NUM_THREADS clause

```
#pragma omp parallel if(np>1) num_threads(np)
{
    …
}
```

- **Execute** in **parallel** **if expression is true**
- **Executes using *np* number of threads**

omp_set_num_threads( ) function

```
#define TOTAL_THREADS 8
int main( )
{
    omp_set_num_threads(TOTAL_THREADS);
    #pragma omp parallel
    {
        . . .
    }
. . .
```

- Execute in parallel <u>using 8 threads</u>

OMP_NUM_THREADS – Environment Variable

```
$ export OMP_NUM_THREADS=4
$ echo $OMP_NUM_THREADS
```

- **Sets** and **displays** the **value** of the **environment variable** **OMP_NUM_THREADS**

Execution Status in Parallel Region

```
int omp_in_parallel()
```

- Returns non-zero: if execution is in parallel region
- Returns zero: if execution in non-parallel region

Demo: PRegion.c

Shared and Private Data

- **Shared data** are **accessible** by **all threads**

- A **reference** **a[5]** to a **shared array** **accesses** the **same address in all threads**

- **Private data** are **accessible only** by **a thread**
  - **Each thread** has its **own copy**

- The **default** is **shared**

Shared and Private Data

```c
int main(int argc, char* argv[])
{
    int threadData = 10;

    // Beginning of parallel region
    #pragma omp parallel private(threadData)
    {
        threadData =200;
    }

    // Ending of parallel region
    printf("Value: %d\n", threadData);
}
```

Demo: SPData.c

Shared and Private Data

```
#pragma omp parallel shared(list)
```

- **Default behavior**
- **List will be shared**
- **Each thread access the same memory location**
- **Initial value (<u>for the first thread</u>) will be same as before the region**
- **Final value will be updated by the <u>last thread</u> leaving the region**
- **Problems: <u>Data Race</u>**

Shared and Private Data

```
#pragma omp parallel private (list)
```

- **Data local to thread**
- You **should not rely** on any **initial and terminal value (after execution of the parallel region)**
- **Separate "Stack Memory" for each thread's private data**
- **No storage associated** with **original object (even with same name for data-items)**

- Use ***firstprivate*** and/or ***lastprivate*** ***clause*** to override

Shared and Private Data

```
#pragma omp parallel firstprivate (list)
```

- **Variables** in **list** are **private**
- **Initialized** with the **value** the **variable had** ***before*** **entering** the **construct**

```
#pragma omp parallel for lastprivate (list)
```

- **Used in “for” loops**
- **Variables** in **list** are **private**
- The **thread** that **executes** the ***final iteration of the loop*** **updates** the **value** (of the variables in the list)

Shared and Private Data

```
#pragma omp parallel default (private) shared(list)
#pragma omp parallel default (shared) private(list)
#pragma omp parallel default (none) private(list) shared(list)
```

- **Alter the default behavior**
- **To implement customized access behavior**

Shared and Private Data – Example (1/4)

```
#include <omp.h>
#include <stdio.h>
int main() {
  int i;
  const int N = 1000;
  int a = 50;
  int b = 0;

#pragma omp parallel for default(shared)
  for (i=0; i<N; i++) {
    b = a + i;
  }

  printf("a=%d b=%d (expected a=50 b=1049)\n", a, b);
}
```

Demo: SPDE1.c

Shared and Private Data – Example (2/4)

```c
#include <omp.h>
#include <stdio.h>
int main() {
  int i;
  const int N = 1000;
  int a = 50;
  int b = 0;

#pragma omp parallel for default(none) private(i) private(a) private(b)
  for (i=0; i<N; i++) {
    b = a + i;
  }

  printf("a=%d b=%d (expected a=50 b=1049)\n", a, b);
}
```

Demo: SPDE1.c

Shared and Private Data – Example (3/4)

```c
#include <omp.h>
#include <stdio.h>
int main() {
  int i;
  const int N = 1000;
  int a = 50;
  int b = 0;

#pragma omp parallel for default(none) private(i) private(a) lastprivate(b)
  for (i=0; i<N; i++) {
    b = a + i;
  }

  printf("a=%d b=%d (expected a=50 b=1049)\n", a, b);
}
```

Demo: SPDE1.c

Shared and Private Data – Example (4/4)

```
#include <omp.h>
#include <stdio.h>
int main() {
  int i;
  const int N = 1000;
  int a = 50;
  int b = 0;

#pragma omp parallel for default(none) private(i) firstprivate(a) lastprivate(b)
  for (i=0; i<N; i++) {
    b = a + i;
  }

  printf("a=%d b=%d (expected a=50 b=1049)\n", a, b);
}
```

Demo: SPDE1.c

Getting ID of Current Thread

```
int main(int argc, char* argv[])
{
  int iam, nthreads;
  #pragma omp parallel private(iam,nthreads) num_threads(2)
  {
      iam = omp_get_thread_num();
      nthreads = omp_get_num_threads();
      printf("ThradID %d, out of %d threads\n", iam, nthreads);
      if (iam == 0)
          printf("Here is the Master Thread.\n");
      else
          printf("Here is another thread.\n");
   }

}
```

Demo: CTID.c

Work-Sharing Constructs

- If **all** the **threads** are **doing** the **same thing**, ***what is the advantage then?***

- **Within** each "**Team**" **threads are assigned IDs**, with **master thread** assigned **ID 0**
  - `omp_get_thread_num() //to get thread number`

Can we use this to distribute tasks amongst the "team" members?

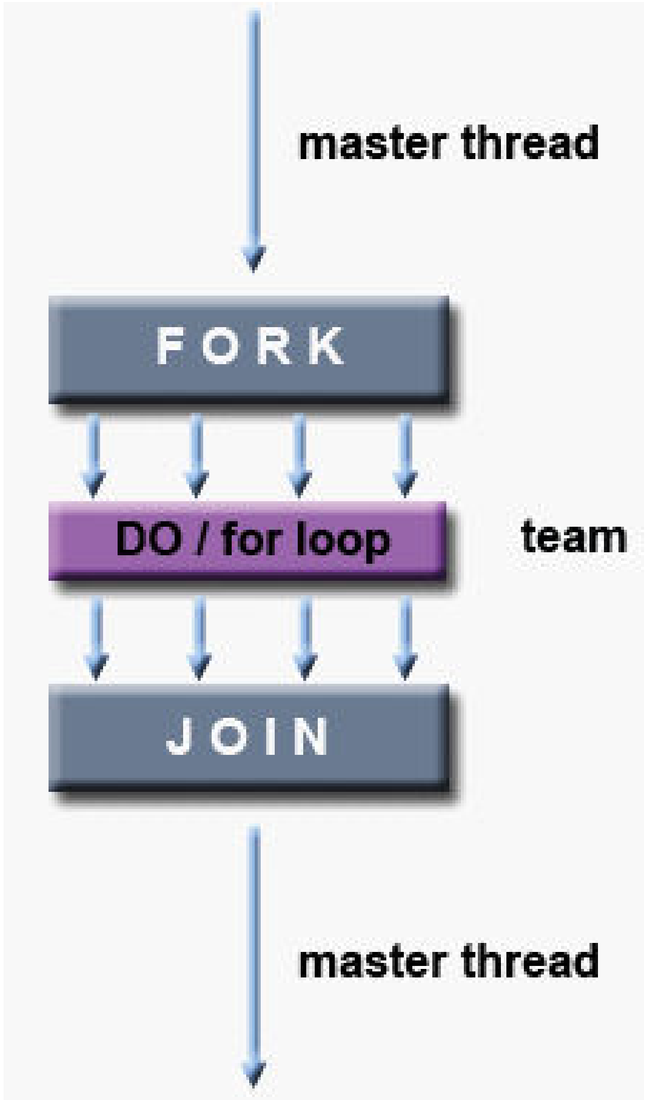
- **Work-sharing constructs distribute** the **specified work** to **all threads** within the **current team**

- **Work-sharing constructs do not launch new threads**

Do/For Work-Sharing Construct

- **DO** / **for** - <u>shares iterations</u> of a **loop across the team**

#pragma omp for [clause ...] newline



There is an implicit synchronization after *#pragma omp for*

Do/For Work-Sharing Construct

- **SCHEDULE clause describes how iterations** of the **loop** are **divided among the threads** in the team

Chunks of specified size assigned round-robin

Chunks of specified size are assigned when thread finishes previous chunk (*work-Stealing mechanism*)

Do/For Work-Sharing Construct

```
int main(int argc, char* argv[])
{
int i, a[10];
#pragma omp parallel num_threads(2)
{
        #pragma omp for schedule(static, 2)
                for ( i=0; i<10;i++)
                        a[i] = omp_get_thread_num();
}

for ( i=0; i<10;i++)
        printf("%d",a[i]);
}
```

Demo: ForConst.c

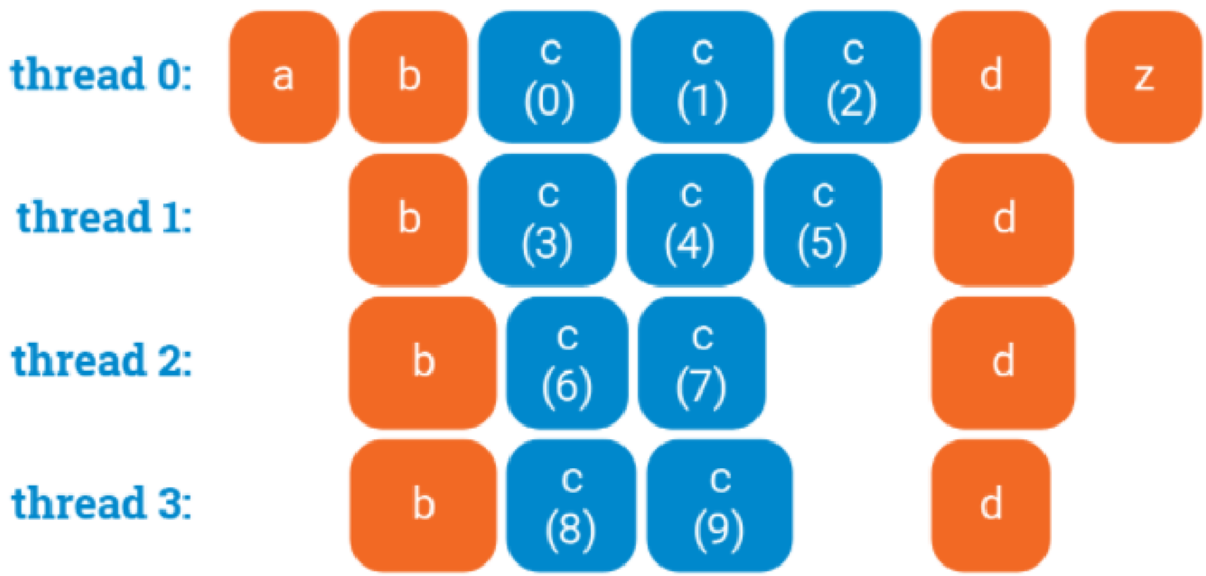
Do/For Work-Sharing Construct

```c
int main(int argc, char* argv[])
{
    int sum, counter, inputList[6] = {11,45,3,5,12,-3};
    #pragma omp parallel num_threads(2)
    {
       #pragma omp for schedule(static, 3)
      for (counter=0; counter<6; counter++) {
     printf("%d adding %d to the 
sum\n",omp_get_thread_num(), inputList[counter]);

     sum+=inputList[counter];
       } //end of for
    } //end of parallel section

    printf("The summed up Value: %d", sum);
}
```

Demo: ForConst2.c

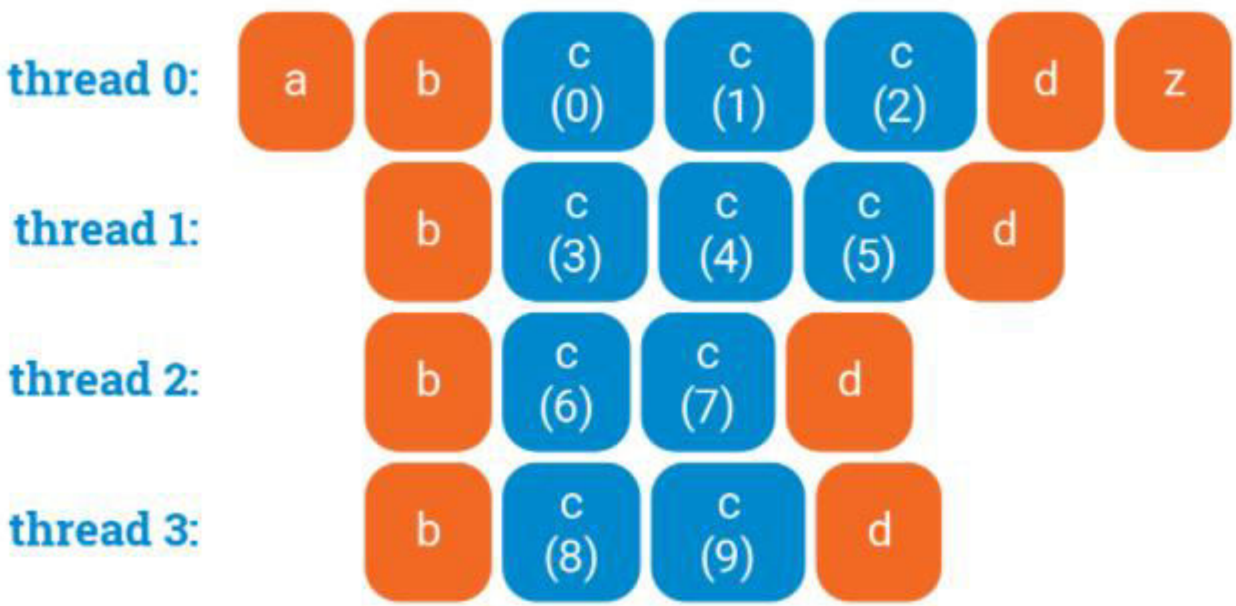
For Work-Sharing –Synchronized

```
a();
#pragma omp parallel
{
    b();
    #pragma omp for
    for (int i = 0; i < 10; ++i) {
        c(i);
    }
    d();
}
z();
```



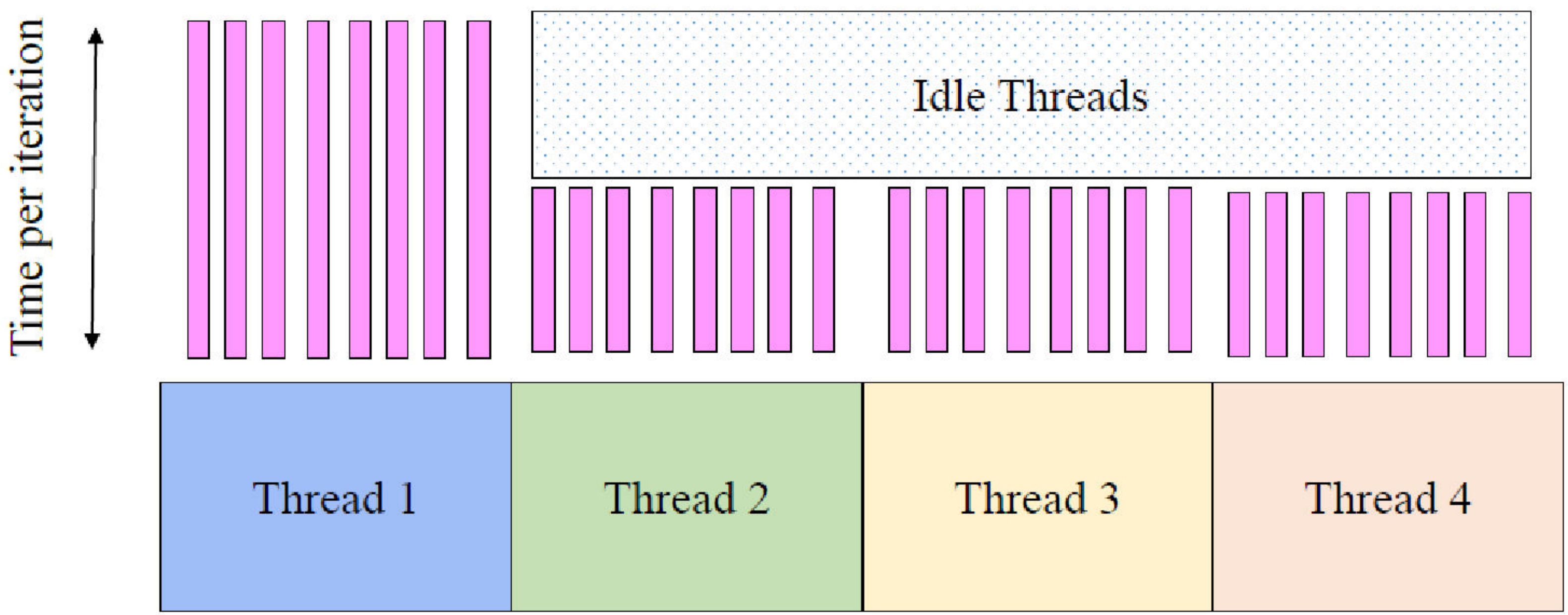
For Work-Sharing – Non Synchronized

```
a();
#pragma omp parallel
{
    b();
    #pragma omp for nowait
    for (int i = 0; i < 10; ++i) {
        c(i);
    }
    d();
}
z();
```



Problems with Static Scheduling

- **What happens if** loop **iterations do not take** the **same** amount of **time**?
  - **Load imbalance**

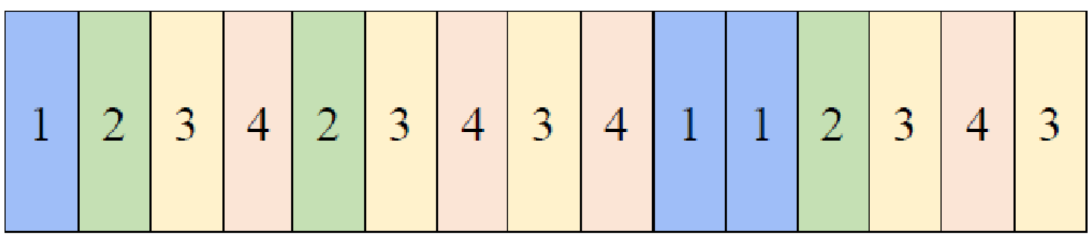


Dynamic Scheduling

- Fixed size chunks assigned on the fly
  - Work-stealing mechanism

- Disadvantage: more overhead as compared to Static

```
#pragma omp parallel for default(shared) private(j) schedule(dynamic,10)
  for (j=0; j<N; j++) {
        ... // some work here
}
```



Demo: LoopSched.c

Threads share Global variables!

```
#include <pthread.h>
#include <iostream>
#include <unistd.h>
using namespace std;
#define NUM_THREADS 10

int sharedData = 0;
void* incrementData(void* arg) {
     sharedData++;
     pthread_exit(NULL);
}

int main()
{
    pthread_t threadID;
    for (int counter=0; counter<NUM_THREADS;counter++) {
         pthread_create(&threadID, NULL, incrementData, NULL);
    }
    cout << "ThreadCount:" << sharedData <<endl;
    pthread_exit(NULL);
}
```

The output for the pthread version?

```
>./globalData
ThreadCount:10

>./globalData
ThreadCount:8
```

ThreadCount: A better implementation

```
#include <pthread.h>
#include <iostream>
#include <unistd.h>
using namespace std;
#define NUM_THREADS 100
int sharedData = 0;
void* incrementData(void* arg)
{
    sharedData++;
    pthread_exit(NULL); }

int main()
{
   pthread_t threadID[NUM_THREADS];
   for (int counter=0; counter<NUM_THREADS;counter++) {
     pthread_create(&threadID[counter], NULL, incrementData, NULL);
   }
   //waiting for all threads
   int statusReturned;
   for (int counter=0; counter<NUM_THREADS;counter++) {
     pthread_join(threadID[counter], NULL);
    }
   cout << "ThreadCount:" << sharedData <<endl;
   pthread_exit(NULL);
}
```

Is the problem solved?

- **Unfortunately, not yet :(**
- The **output** from **running it with 1000 threads** is as below:

```
>./6join
ThreadCount:990
>./6join
ThreadCount:978
>./6join
ThreadCount:1000
>
```

- **Reasons?**
- **What can be done?**

ThreadCount: OpenMP Implementation

```c
int main(int argc, char* argv[])
{
  int threadCount=0;
  #pragma omp parallel num_threads(100)
  {
    int myLocalCount = threadCount;
    sleep(1);
    myLocalCount++;
    threadCount = myLocalCount;


  }
  printf("Total Number of Threads: %d\n", threadCount);
}
```

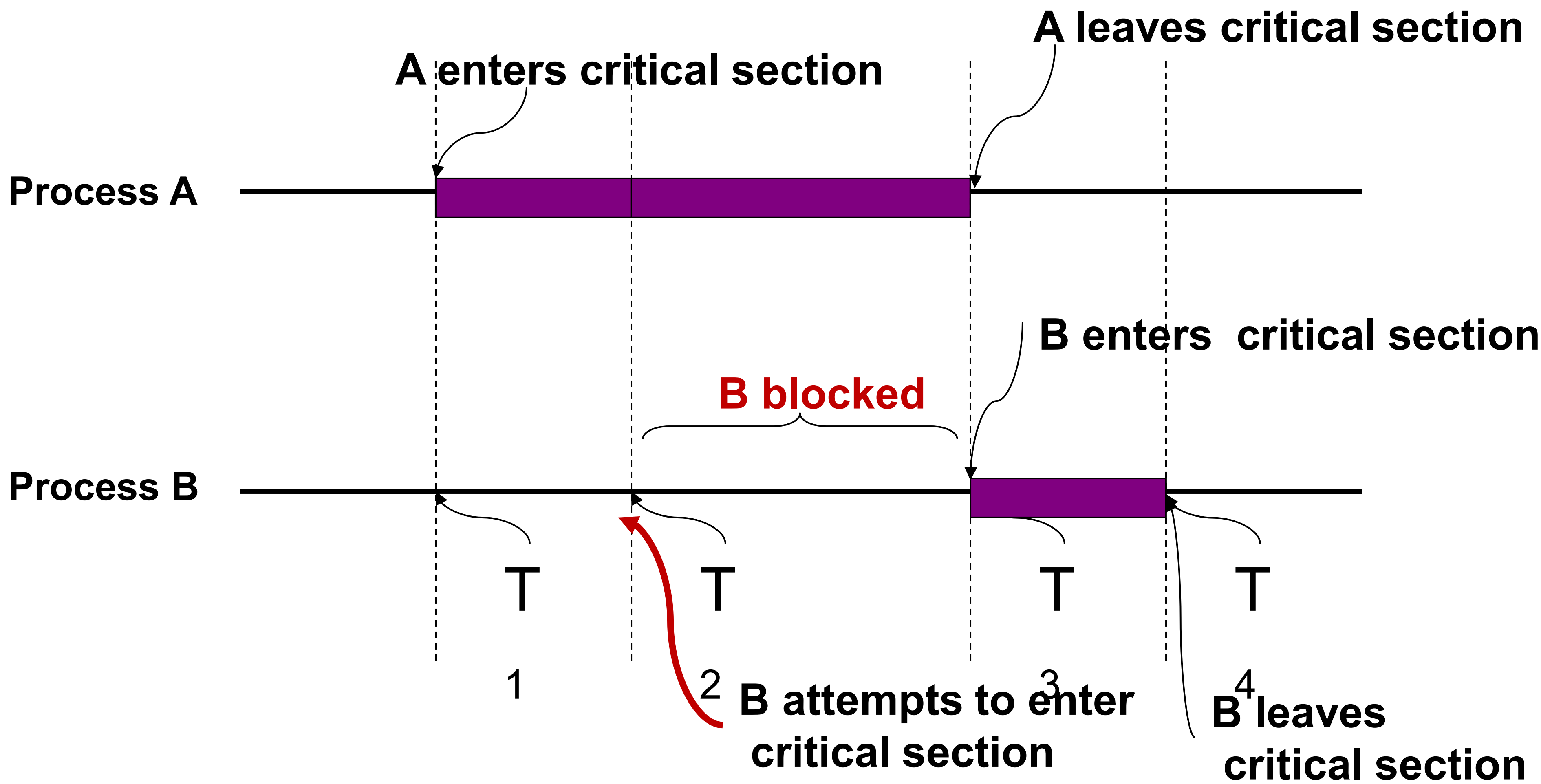
Demo: TCount1.c

Critical-Section (CS) Problem

- ***n*** **processes** all **competing** to **use** some **shared data**
- **Each process** has a **code segment**, called **critical section**, in which the **shared data is accessed**

- **Problem (ensures that):**
  – **Two process** are **not allowed** to **execute** in their **critical section at the same time**
  – **Access** to the **critical section** must be an **atomic action**

Critical Section



Mutual Exclusion

At any given time, only one process is in the critical section

...back to threads counting

```
int sharedData = 0;
pthread_mutex_t mutexIncrement;

void* incrementData(void* arg)
{
    pthread_mutex_lock(&mutexIncrement);
    sharedData++;
    pthread_mutex_unlock(&mutexIncrement);
    pthread_exit(NULL);
}

int main()
{
  pthread_mutex_init(&mutexIncrement, NULL);


  pthread_t threadID[NUM_THREADS];
   for (int counter=0; counter<NUM_THREADS;counter++) {
      pthread_create(&threadID[counter], NULL, incrementData, NULL);
   }
   //waiting for all threads
   int statusReturned;
   for (int counter=0; counter<NUM_THREADS;counter++) {
      pthread_join(threadID[counter], NULL);
    }
   cout << "ThreadCount:" << sharedData <<endl;
   pthread_exit(NULL);
}
```

OpenMP - Synchronization Constructs

- The **CRITICAL directive specifies** a **region of code** that **must** be **executed** by **only one thread at a time**

- If a **thread** is **currently executing** inside a **CRITICAL region** and **another thread attempts** to **execute it**, **it will block until the first thread exits** that CRITICAL region.

```
pragma omp critical [ name ]
...
```

... back to threadCount

```
int main(int argc, char* argv[])
{
    int threadCount;
    #pragma omp parallel num_threads(5)
    {
        #pragma omp critical
        {
            int myLocalCount = threadCount;
            sleep(1);
            myLocalCount++;
            threadCount = myLocalCount;
        }
    }
    printf("Total Number of Threads: %d\n", threadCount);
}
```

Demo: TCount2.c

OpenMP - Synchronization Constructs

- The **MASTER directive specifies** a **region** that is to be **executed only by the master thread** of the **team**

- **All other threads** on the team **skip this section** of **code**

```
#pragma omp master
...
```

Demo: MasterOnly.c

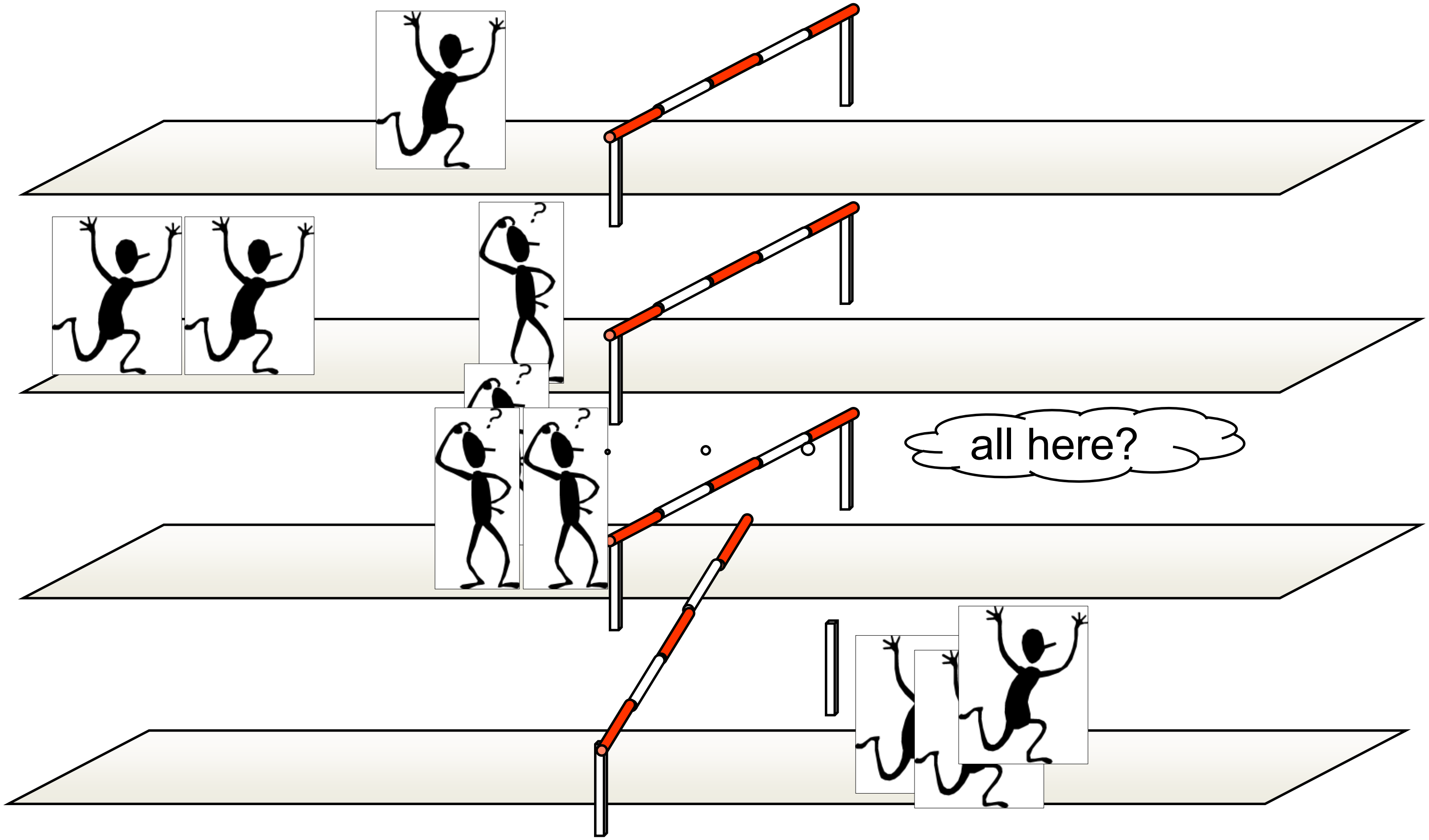
OpenMP - Synchronization Constructs

- When a **BARRIER** **directive** is **reached**, a **thread will wait** at that **point** **until** all other **threads have reached that barrier**

- ***All threads then resume executing in parallel the code that follows the barrier.***

```
#pragma omp barrier
...
```

Barrier Synchronization



Demo: Barrier.c

Reduction (Data-sharing Attribute Clause)

- The **REDUCTION** **clause** **performs** a **reduction** **operation** on the **variables** that **appear in the list**
- A **private copy** for **each list variable** is **created** and **initialized** for **each thread**
- At the **end of the reduction**, the **reduction variable (all private copies)** **is examined and the shared variable's final result** is **written.**

#pragma omp operator: list

...

operator can be `+,-,*,&&,||,max,min ...`

Reduction (Data-sharing Attribute Clause)

```
int main(int argc, char* argv[])
{
    srand(time(NULL));
    int winner;
    #pragma omp parallel reduction(max:winner) num_threads(10)
    {
        winner = (rand() % 1000) + omp_get_thread_num();
        printf("Thread: %d has Chosen: %d\n", 
omp_get_thread_num(),winner);
    }
    printf("Winner: %d\n", winner);

}
```

Demo: Reduction.c

Any Questions?